

BHOOMIKA RAVISHANKAR

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Information Management

Honors & Awards: Phi Kappa Phi Honor Society (Top 10% of Graduate Students)

August 2024 - May 2026

GPA: 4.0 /4.0

Savitribai Phule Pune University

Bachelor of Engineering in Computer Engineering

August 2020 - June 2024

GPA: 9.12 /10.0

TECHNICAL SKILLS

Programming & Data Analytics/BI: Python, SQL, R, SAS, Pandas, NumPy, Scikit-learn, Git, Tableau, Power BI, A/B Testing, KPI, MBR/QBR

Data Engineering & Cloud: Spark, AWS S3, EMR, Glue, Azure, GCP, PySpark, Airflow, Databricks, Snowflake, BigQuery, Hadoop, Hive

Machine Learning & Statistics: Logistic Regression, XGBoost, Random Forest, LSTM, Time Series Forecasting, PD/EAD, CECL, Risk Management, Stress Testing, Model Validation, Feature Engineering, SMOTE, Lasso/Ridge, Causal Inference, Clustering, Data Storytelling

NLP & GenAI: LLM Fine-tuning, RAG, AgenticAI, Hugging Face Transformers, Sentence Embeddings, Entity Extraction, TensorFlow, PyTorch

WORK EXPERIENCE

Synchrony Financial, Champaign, IL

May 2025 - May 2026

Data Scientist, Credit Risk Division

- Developed and deployed statistical **PD/EAD models** to production via AWS SageMaker within Synchrony's AtLAS platform, a complementary log-log GLM framework predicting **5-year** charge-off probability across **6** retail portfolios (clothing, lawn & garden, lifestyle, luxury, sporting goods) directly informing Synchrony's **\$7.49B CECL loss reserve**.
- Engineered **13+ longitudinal features** spanning credit bureau scores, utilization groups, and behavioral interactions alongside macroeconomic signals (unemployment lag, GDP, savings rate, HPI, median income) across **15+ years** of snapshots; retained variables drove **75%** deviance reduction from the null model with **near-zero p-values** (hypothesis test) and stable **VIF ≤ 3** across interaction terms.
- Validated model robustness through multi-snapshot backtesting from **2007–2025**, confirming conservative under-prediction (**~10%**) pre-2008 for regulatory compliance, self-correction within **~1 year** following the 2008 financial shock, and stable charge-off variance across **S1–S4 stress** periods including post-COVID behavioral shifts.
- Built large-scale PySpark ETL pipelines on AWS EMR, scheduled via Apache Airflow, to ingest and transform **900M+** customer records per snapshot, reducing pipeline runtime from **1 hr (Python) to 5 mins (PySpark)**, **92%** faster.

Data Scientist, Operations Analytics Division

- Replaced an external consulting firm's NPS model with an internally built solution, **eliminating third-party vendor costs**; findings on top NPS-driving survey questions were adopted by **customer analytics leadership** to redesign survey structure and cut low-signal questions.
- Migrated legacy SAS/SQL workflows to Python, building a scalable ETL pipeline unifying **100M+** account-level records across 11 survey datasets, cutting runtime from **2 hr to 15 mins (87% faster)** and resolving schema mismatches & deduplication issues that blocked modeling.
- Conducted univariate and multivariate analysis across **10+ surveys**; identified heavily skewed NPS targets and **90% sparse survey-level** coverage, triggering a redesign of modeling granularity from account-level to survey-level to improve signal quality and model reliability.
- Modeled NPS drivers at the Apply Experience Survey level using XGBoost (**R2/RMSE similar, confirming no overfitting**) and **Lasso regression (R² 0.43)** in parallel, with mean/median imputation across **100K+** missing values; Lasso coefficients served as a cross-validation signal to identify which survey questions most consistently predicted NPS movement.
- Validated causal robustness using **Placebo Refutation Testing**, replacing the treatment variable with a random placebo and confirming **near-zero effect (-0.001)**, establishing that identified survey drivers reflect genuine signal rather than random correlation.

University of Illinois Urbana-Champaign, Champaign, IL

January 2025 - Present

Research Assistant, AI Engineer

- Developed an end-to-end **LLM-driven NLP pipeline** converting unstructured text from **100,000** clinical trial studies into structured model-ready variables using Qwen 2.5 7B-Instruct, extracting named entities (drug names, conditions, trial phases) with one-hot encoding, sentence embeddings, and PCA to help clinicians predict success of trials.
- Benchmarked LLM-derived features** against sentence embeddings across a Lasso logistic regression evaluation workflow, finding LLM features contributed **97.2%** of total feature importance vs. **2.8%** from embeddings, retaining only 5 of 50 embedding features and directly driving a **full pipeline redesign** toward LLM-first feature extraction.
- Led an iterative **prompt refinement** cycle using **L1-penalized Lasso (0.79 F1, 0.59 5-fold ROC-AUC)** as a feedback signal, identifying underrepresented variables, redesigning extraction prompts, & re-running pipeline to improve **feature quality** for **clinical trial classification**.

Business Intelligence Group (BIG), Champaign, IL

January 2025 - May 2025

Data Science Consultant

- Designed and implemented an end-to-end **NLP decision support system** integrating a **RAG-based Python chatbot**, using **retrieval-augmented generation** for contract query resolution and **topic modelling** for intent classification, with Power BI data visualizations to automate resource allocation and overtime planning, reducing pilot lookup time by **90%**.

PROJECTS

- AI-driven M&A Prediction:** Architected a multi-modal M&A target prediction system combining 14 LSEG financial ratios with **298K+** financial news headlines, using NER to extract company and ticker mentions, **FinBERT sentiment**, **Word2Vec** across **3,882 companies**; applied SMOTE to address **20:1 class imbalance** and XGBoost with 5-fold CV, achieving **89% accuracy** and **2.4× F1 improvement**.
- LSTM Credit Spend Forecasting:** Trained a 2-layer LSTM to forecast **Q4 2025 credit card spend** across 14,099 accounts using **13-month transaction histories**, achieving **\$2,444 MAE**, enabling credit recommendations for 288 high-value accounts (2% segment).
- Ambiguity-Clarifying LLMS:** Fine-tuned Qwen 2.5 70B on an **ambiguity-labeled clarification dataset** to improve question understanding and response specificity, applied DPO and achieved **25% increase** in clarification accuracy across various ambiguity types.